

Ecuaciones diferenciales ordinarias

Para el problema del paracaidista se tiene que

$$\frac{dv(t)}{dt} = g - \frac{c}{m} \left(v(t) + a \left(\frac{v(t)}{v_{\max}} \right)^b \right)$$

$$C = 12.5 \text{ kg/s}$$

$$m = 68.1 \text{ kg}$$

$$a = 8.3 \text{ m/s}$$

$$b = 2.2$$

$$v_{\max} = 46 \text{ m/s}$$

$$g = 9.81 \text{ m/s}$$

Calcular $v(t)$ usando el método de Euler y el R-K de 2° y 4° orden con $h = 0.1 \text{ s}$ y $v(0) = 0$

$$\frac{c}{m} = \frac{12.5}{68.1} = 0.1835 \frac{\text{kg}}{\text{kg/s}}$$

Método de Euler

$$v_{i+1} = v_i + h \cdot f(v_i)$$

★ $t = 0$ a $t = 0.1$

$$v_0 = 0$$

Calculamos $f(v_0)$:

$$\left(\frac{v_0}{v_{\max}} \right)^b = \left(\frac{0}{46} \right)^{2.2} = 0$$

$$f(v_0) = 9.81 \text{ m/s} - 0.1835 \text{ 1/s} [0 + 8.3(0)] = 9.81 \text{ m/s}$$

★ $t = 0.1$ a $t = 0.2$

$$v_1 = 0.981$$

$$\left(\frac{v_1}{v_{\max}} \right)^{2.2} = 0.00004176$$

$$f(v_i) = 9.81 - 0.18355 \left[0.981 + 8.3(0.000417) \right]$$

$$f(v_i) = 9.62984$$

Método de Runge-Kutta de segundo orden

Fórmulas

$$K_1 = f(v_i)$$

$$K_2 = f(v_i + h \cdot K_1)$$

$$v_{i+1} = v_i + \frac{h}{2} (K_1 + K_2)$$

$$v_0 = 0$$

$$K_1 = f(0) = 9.81$$

$$v_{\text{pred}} = v_0 + h \cdot K_1 = 0 + 0.1(9.81) = 0.981$$

$$K_2 = f(0.981) \approx 9.629$$

$$v_1 = 0 + \frac{0.1}{2} (0.981 + 9.62985) = 0.97199$$

$$* v_1^{\text{RK2}} = 0.97119$$

$$\left(\frac{v_1}{v_{\text{max}}} \right)^{2.2} = 4.10 \times 10^{-5}$$

$$v + a \left(\frac{v_1}{v_{\text{max}}} \right)^{2.2} = 0.97199 + 0.00034 = 0.97233$$

$$K_1 = f(v_i) = 9.81 - 0.1835 (0.97233) = 9.6315$$

Predecimos $V_{temp} = V_1 + h \cdot K_1 = 0.97199 + (0.1)(9.6315)$

$$V_{temp} = 1.9351$$

$$K_2 = f(1.935142)$$

$$\left(\frac{V_2}{V_{max}} \right)^{2.2} = \left(\frac{1.9351}{9.6} \right)^{2.2} = 0.0001865$$

$$V + a \left(\frac{V_2}{V_{max}} \right)^{2.2} = 1.9366$$

$$K_2 = 9.81 - 0.1835(1.9366) \approx 9.4545$$

Promediamos $V_2 = V_1 + \frac{h}{2} (K_1 + K_2)$

$$V_2 = 0.97199 + 0.05(9.6315 + 9.4545) = 1.92629$$

Resultados: $V_0 = 0$ $V_1 = 0.972$ $V_2 = 1.9263$

Método de Runge-Kutta de cuarto orden.

$$K_1 = f(v_i)$$

$$K_2 = f\left(v_i + \frac{h}{2} K_1\right)$$

$$K_3 = f\left(v_i + \frac{h}{2} K_2\right)$$

$$K_4 = f(v_i + h K_3)$$

$$V_{i+1} = V_i + \frac{h}{6} (K_1 + 2K_2 + 2K_3 + K_4)$$

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$$K_1 = f(0) = 9.81$$

$$V_0 + \frac{h}{2} K_1 = 0 + (0.05)(9.81) = 0.4905$$

Calculamos $f(0.4905)$

$$V + a \left(\frac{V}{V_{\max}} \right)^{2.2} = 0.49057$$

$$K_2 = 9.81 - 0.1835(0.49057) = 9.7199$$

$$\text{Ahora } K_3 = f\left(0 + \frac{h}{2} K_2\right) = f(0.48599)$$

$$V + a \left(\frac{V}{V_{\max}} \right)^{2.2} = 0.486069$$

$$K_3 = 9.81 - 0.1835(0.486069) = 9.72075$$

$$\text{Ahora } K_4 = f\left(V_0 + h K_3\right) = f(0.972075)$$

$$V + a \left(\frac{V}{V_{\max}} \right)^{2.2} = 0.972415$$

$$K_4 = 9.81 - 0.1835(0.972415) = 9.63145$$

$$V_1 = 0 + \frac{0.1}{6} (9.81 + 2(9.7199) + 2(9.72075) + 9.63145)$$

$$V_1 = 0.97205$$

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$t(s)$	$V_{Euler} (m/s)$	$V_{RK2} (m/s)$	$V_{RK4} (m/s)$
0	0.000	0.000	0
0.1	0.9810	0.9720	0.9721
0.2	1.9440	1.9263	1.9252